DETAIL MINI PROJECT

Area: Blockchain

Mata Kuliah: Komputasi Berbasis Jaringan — Program S2 Tesis

Timeline: 3 Pekan Implementasi + 1 Pekan Pengujian Komprehensif

Dokumen ini memuat detail teknis untuk 4 topik mini project, masing-masing mencakup deskripsi, infrastruktur, langkah pengerjaan, dan skenario uji coba komprehensif beserta template pengumpulan data.

TOPIK 3A

Desain dan Evaluasi Reputation-Based Consensus untuk Permissioned Blockchain pada Skenario Supply Chain

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1. Deskripsi

Mekanisme konsensus di mana hak validasi ditentukan oleh skor reputasi node. Node yang sering mengajukan transaksi invalid kehilangan hak voting secara gradual. Custom lightweight blockchain Python. Perbandingan 4 mekanisme: PBFT (baseline), Proof-of-Authority (PoA), simplified DPoS, dan Reputation-Based (proposal). Bisa jalan di Google Colab tanpa Docker.

2. Infrastruktur

- 7 Python processes (5 honest + 2 byzantine) via socket localhost.
- Blockchain Core (~300 LOC), Consensus Module (~200 LOC, 4 implementasi), Reputation Manager (~100 LOC).
- Byzantine Simulator: 3 behavior (invalid TX, inconsistent vote, delay). Transaction Generator: sebagian sengaja invalid.

3. Langkah

Pekan 1: Blockchain core + PBFT + process setup. Pekan 2: Reputation + PoA + DPoS + Byzantine. Pekan 3: Integrasi + tuning. Pekan 4: Eksekusi + report.

4. Skenario Uji Coba dan Matriks Evaluasi

4.1 Variabel Tetap

- 7 nodes, block size 50 tx, block interval 5 detik, EMA α=0.9, threshold=0.3, seed tetap, warm-up 10 block pertama di-skip, genesis reset antar run, TX rate: 50 tx/s.

4.2 Variabel Manipulasi

Variabel	Level
Konsensus	PBFT, PoA, DPoS, Reputation-Based
Jumlah Byzantine Nodes	0 (all honest), 1 node, 2 nodes
Byzantine Behavior	Invalid-TX-Only, Inconsistent-Vote, Mixed (keduanya)
Invalid TX Rate	0%, 10%, 30%

Total: 4 konsensus × 3 byzantine count × 3 behavior × 3 invalid rate = 108 kombinasi. Namun 0 byzantine tidak terpengaruh behavior dan invalid rate, sehingga efektif = (4×1) + (4×2×3×3) = 4 + 72 = 76 × 5 rep = 380 run. Prioritas: 2 byzantine + Mixed behavior = 4 × 3 invalid rate = 12 × 5 = 60 run (~10 jam).

4.3 Variabel Respon

Variabel	Satuan	Pengukuran
Throughput (TPS)	tx/s	Transaksi valid committed per detik

Consensus Latency	ms	Dari proposal hingga finality
Block Finality Rate	%	Block finalized / total proposed
Byzantine Detection Rate	%	Byzantine node yang reputasinya < threshold
False Positive Rate	%	Honest node yang turun salah
Reputation Convergence	blocks	Block hingga byzantine teridentifikasi
Chain Consistency	boolean	Semua honest node chain identik
Message Complexity	msg/block	Total pesan konsensus per block

4.4 Template Pengumpulan Data

Isi setiap sel dengan rata-rata ± standar deviasi dari 5 repetisi.

Tabel 4.4a — Byzantine: 0 nodes (All Honest), Invalid TX Rate: 0%

Metrik	PBFT	PoA	DPoS	Reputation
TPS (tx/s)	___±___	___±___	___±___	___±___
Consensus Latency (ms)	___±___	___±___	___±___	___±___
Finality Rate (%)	___±___	___±___	___±___	___±___
Chain Consistent	___	___	___	___
Msg/Block	___±___	___±___	___±___	___±___

Tabel 4.4b — Byzantine: 1 node, Behavior: Invalid-TX-Only, Invalid Rate: 10%

Metrik	PBFT	PoA	DPoS	Reputation
TPS (tx/s)	___±___	___±___	___±___	___±___
Consensus Latency (ms)	___±___	___±___	___±___	___±___
Finality Rate (%)	___±___	___±___	___±___	___±___
Detection Rate (%)	—	—	—	___±___
False Positive (%)	—	—	—	___±___
Convergence (blocks)	—	—	—	___±___
Chain Consistent	___	___	___	___
Msg/Block	___±___	___±___	___±___	___±___

Tabel 4.4c — Byzantine: 1 node, Behavior: Invalid-TX-Only, Invalid Rate: 30%

Metrik	PBFT	PoA	DPoS	Reputation
TPS (tx/s)	___±___	___±___	___±___	___±___
Consensus Latency (ms)	___±___	___±___	___±___	___±___
Finality Rate (%)	___±___	___±___	___±___	___±___
Detection Rate (%)	—	—	—	___±___
Convergence (blocks)	—	—	—	___±___
Chain Consistent	___	___	___	___

Tabel 4.4d — Byzantine: 2 nodes, Behavior: Invalid-TX-Only, Invalid Rate: 10%

Metrik	PBFT	PoA	DPoS	Reputation
TPS (tx/s)	___±___	___±___	___±___	___±___

Consensus Latency (ms)	___±___	___±___	___±___	___±___
Finality Rate (%)	___±___	___±___	___±___	___±___
Detection Rate (%)	—	—	—	___±___
False Positive (%)	—	—	—	___±___
Convergence (blocks)	—	—	—	___±___
Chain Consistent	___	___	___	___
Msg/Block	___±___	___±___	___±___	___±___

Tabel 4.4e — Byzantine: 2 nodes, Behavior: Invalid-TX-Only, Invalid Rate: 30%

Metrik	PBFT	PoA	DPoS	Reputation
TPS (tx/s)	___±___	___±___	___±___	___±___
Finality Rate (%)	___±___	___±___	___±___	___±___
Detection Rate (%)	—	—	—	___±___
Convergence (blocks)	—	—	—	___±___
Chain Consistent	___	___	___	___

Tabel 4.4f — Byzantine: 2 nodes, Behavior: Inconsistent-Vote, Invalid Rate: 0%

Metrik	PBFT	PoA	DPoS	Reputation
TPS (tx/s)	___±___	___±___	___±___	___±___
Consensus Latency (ms)	___±___	___±___	___±___	___±___
Finality Rate (%)	___±___	___±___	___±___	___±___
Detection Rate (%)	—	—	—	___±___
Convergence (blocks)	—	—	—	___±___
Chain Consistent	___	___	___	___

Tabel 4.4g — Byzantine: 2 nodes, Behavior: Mixed (Invalid-TX + Inconsistent-Vote), Invalid Rate: 10%

Metrik	PBFT	PoA	DPoS	Reputation
TPS (tx/s)	___±___	___±___	___±___	___±___
Consensus Latency (ms)	___±___	___±___	___±___	___±___
Finality Rate (%)	___±___	___±___	___±___	___±___
Detection Rate (%)	—	—	—	___±___
False Positive (%)	—	—	—	___±___
Convergence (blocks)	—	—	—	___±___
Chain Consistent	___	___	___	___
Msg/Block	___±___	___±___	___±___	___±___

Tabel 4.4h — Byzantine: 2 nodes, Behavior: Mixed, Invalid Rate: 30% (WORST CASE)

Metrik	PBFT	PoA	DPoS	Reputation
TPS (tx/s)	___±___	___±___	___±___	___±___
Consensus Latency (ms)	___±___	___±___	___±___	___±___
Finality Rate (%)	___±___	___±___	___±___	___±___

Detection Rate (%)	—	—	—	___±___
False Positive (%)	—	—	—	___±___
Convergence (blocks)	—	—	—	___±___
Chain Consistent	___	___	___	___
Msg/Block	___±___	___±___	___±___	___±___

4.5 Uji Statistik

- Mann-Whitney U per pasangan konsensus. Kruskal-Wallis: 4 sekaligus.
- Survival analysis (Kaplan-Meier): time-to-detection byzantine per mechanism.
- Chi-square: chain consistency rate antar konsensus.
- Sensitivity: EMA α = {0.7, 0.8, 0.9, 0.95} pada Reputation + 2 byz + Mixed 30%.

4.6 Visualisasi

- Time-series reputation score per node (1 plot per konsensus pada worst case 4.4h).
- Bar chart TPS + finality per konsensus, grouped per byzantine scenario.
- Line chart validator set size over blocks.
- Kaplan-Meier curve: time-to-detection per mechanism.

TOPIK 3B

Gas Optimization Framework untuk Smart Contract Solidity Menggunakan Static Analysis

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1. Deskripsi

Tool analisis statik yang mendeteksi 6 anti-pattern boros gas pada Solidity, memberikan rekomendasi refactoring, dan mengukur penghematan gas empiris. Diuji pada 50–100 contracts dari Etherscan. Head-to-head vs Slither.

2. Infrastruktur

- Python AST parser + solc + Hardhat in-memory testnet. Dataset: 50–100 verified contracts, stratified 10 per domain.
- Gas 100% deterministik di Hardhat. Bisa di Colab.

3. Langkah

Pekan 1: solc + Hardhat + download contracts + AST parser. Pekan 2: 6 pattern detectors + gas estimator. Pekan 3: Refactoring recommender + Slither comparison. Pekan 4: Full analysis + report.

4. Skenario Uji Coba

4.1 Variabel Tetap

- solc 0.8.20, Hardhat network, optimization off, gas measurement 3x median, manual audit 20 contracts (2 reviewer, Cohen's kappa).

4.2 Variabel Manipulasi

Variabel	Level
Anti-Pattern Type	Redundant SLOAD, Unoptimized Loop, String vs Bytes32, Public vs External, Unchecked Arithmetic, Dead Code
Contract Domain	DeFi, NFT, Token (ERC-20/721), Governance, Utility
Contract Complexity	Simple (<100 LOC), Medium (100–500 LOC), Complex (500+ LOC)

4.3 Variabel Respon

Variabel	Satuan	Pengukuran
Precision (per pattern)	%	Correctly detected / total detected
Recall (per pattern)	%	Detected / actual (manual audit)
Gas Saved per Pattern	gas	Original vs refactored
Gas Saved Percentage	%	(Saved / original) × 100
Auto-Refactor Success	%	Compiles + tests pass
Analysis Time	detik	Wall-clock per contract
False Positive Rate	%	Incorrect / total detected

4.4 Template Pengumpulan Data

Tabel 4.4a — Detection Accuracy per Anti-Pattern (dari 20 manually-audited contracts)

Anti-Pattern	True Pos	False Pos	False Neg	Precision (%)	Recall (%)	Contracts Affected
Redundant SLOAD	___	___	___	___	___	___/20
Unoptimized Loop	___	___	___	___	___	___/20
String vs Bytes32	___	___	___	___	___	___/20
Public vs External	___	___	___	___	___	___/20
Unchecked Arithmetic	___	___	___	___	___	___/20
Dead Code	___	___	___	___	___	___/20

Tabel 4.4b — Gas Savings per Anti-Pattern (dari 50 contracts, Hardhat measurement)

Anti-Pattern	Contracts w/ Pattern	Avg Gas Before	Avg Gas After	Avg Saved (%)	Max Saved (%)	Refactor Success (%)
Redundant SLOAD	___/50	___	___	___±___	___	___
Unoptimized Loop	___/50	___	___	___±___	___	___
String vs Bytes32	___/50	___	___	___±___	___	___
Public vs External	___/50	___	___	___±___	___	___
Unchecked Arithmetic	___/50	___	___	___±___	___	___
Dead Code	___/50	___	___	___±___	___	___

Tabel 4.4c — Cross-Domain Analysis: Pattern Distribution per Domain (50 contracts)

Anti-Pattern	DeFi (10)	NFT (10)	Token (10)	Governance (10)	Utility (10)
Redundant SLOAD	___	___	___	___	___
Unoptimized Loop	___	___	___	___	___
String vs Bytes32	___	___	___	___	___
Public vs External	___	___	___	___	___
Unchecked Arithmetic	___	___	___	___	___
Dead Code	___	___	___	___	___
Total Patterns	___	___	___	___	___
Avg Gas Saved (%)	___	___	___	___	___

Tabel 4.4d — Complexity Scalability: Akurasi per Complexity Level

Metrik	Simple (<100 LOC)	Medium (100–500)	Complex (500+)
Avg Patterns Detected	___±___	___±___	___±___
Overall Precision (%)	___	___	___
Overall Recall (%)	___	___	___
Analysis Time (s)	___±___	___±___	___±___
False Positive Rate (%)	___	___	___

Tabel 4.4e — Head-to-Head vs Slither (50 contracts)

Metrik	Our Tool	Slither	Overlap
Total Patterns Detected	___	___	___
Unique to Our Tool	___	—	—
Unique to Slither	—	___	—
Precision (on shared patterns)	___	___	—
Gas Quantification	Ya (per pattern)	Tidak	—
Avg Analysis Time (s)	___	___	—

4.5 Uji Statistik

- Cohen’s kappa: inter-rater agreement manual audit.
- McNemar’s test: our tool vs Slither per contract (paired binary).
- Wilcoxon signed-rank: gas before vs after per contract.
- Kruskal-Wallis: gas savings across 5 domains.
- Spearman correlation: LOC vs total patterns detected.

4.6 Visualisasi

- Bar chart gas savings per pattern type. Venn diagram detection overlap vs Slither.
- Heatmap: pattern frequency per domain. Scatter: LOC vs patterns detected.

TOPIK 3C

Analisis Performa State Channel untuk Transaksi Mikro pada Ethereum Layer-2

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1. Deskripsi

State channel on-chain/off-chain untuk micro-TX. Hardhat testnet. Fokus: break-even point, dispute cost, settlement frequency impact. Perbandingan: All On-Chain, State Channel, Multi-Hop Channel.

2. Infrastruktur

- Solidity (~150 LOC) + off-chain client (Node.js/Python ~200 LOC) + Hardhat in-memory. Gas 100% deterministik — 3 repetisi cukup. Total: ~48 menit.

3. Langkah

Pekan 1: Smart contract + Hardhat + tests. Pekan 2: Off-chain client + settlement + dispute. Pekan 3: Gas metering + break-even + multi-hop. Pekan 4: Experiment + report.

4. Skenario Uji Coba

4.1 Variabel Tetap

- Hardhat, solc 0.8.20, 2 participants, TX amount uniform 0.001–0.1 ETH, fixed mnemonic.

4.2 Variabel Manipulasi

Variabel	Level
Mode Transaksi	All On-Chain (baseline), State Channel, Multi-Hop Channel
Jumlah TX per Channel	10, 50, 100, 500, 1000
Settlement Frequency	Every 10 tx, Every 50 tx, At close only
Dispute Scenario	No dispute, 1 dispute mid-channel, Dispute at close

Total: 3 mode × 5 TX count × 3 settlement × 3 dispute = 135 kombinasi. Namun On-Chain tidak terpengaruh settlement/dispute. Efektif: 5 + (2×5×3×3) = 5 + 90 = 95 × 3 rep = 285 run. Karena deterministik dan per run ~10 detik, total ~48 menit.

4.3 Variabel Respon

Variabel	Satuan	Pengukuran
Total Gas Cost	gas	Aggregate gas seluruh lifecycle
Gas per Micro-TX	gas/tx	Total gas / jumlah TX
Off-Chain Throughput	tx/s	TX per detik off-chain
Break-Even Point	tx count	Min TX di mana channel lebih murah
Dispute Gas Cost	gas	Extra gas untuk dispute
Settlement Overhead	gas	Gas untuk periodic settlement

Channel Open/Close Gas	gas	Fixed cost

4.4 Template Pengumpulan Data

Tabel 4.4a — Settlement: At Close Only, Dispute: No Dispute

TX Count	Gas On-Chain	Gas Channel	Gas Multi-Hop	Gas/TX On-Chain	Gas/TX Channel	Break-Even?
10	___	___	___	___	___	Ya/Tidak
50	___	___	___	___	___	Ya/Tidak
100	___	___	___	___	___	Ya/Tidak
500	___	___	___	___	___	Ya/Tidak
1000	___	___	___	___	___	Ya/Tidak

Tabel 4.4b — Settlement: Every 10 TX, Dispute: No Dispute

TX Count	Gas On-Chain	Gas Channel	Gas Multi-Hop	Settlement Overhead	Break-Even?
10	___	___	___	___	Ya/Tidak
50	___	___	___	___	Ya/Tidak
100	___	___	___	___	Ya/Tidak
500	___	___	___	___	Ya/Tidak
1000	___	___	___	___	Ya/Tidak

Tabel 4.4c — Settlement: Every 50 TX, Dispute: No Dispute

TX Count	Gas On-Chain	Gas Channel	Gas Multi-Hop	Settlement Overhead	Break-Even?
50	___	___	___	___	Ya/Tidak
100	___	___	___	___	Ya/Tidak
500	___	___	___	___	Ya/Tidak
1000	___	___	___	___	Ya/Tidak

Tabel 4.4d — Settlement: At Close Only, Dispute: 1 Dispute Mid-Channel

TX Count	Gas No-Dispute	Gas w/ Dispute	Dispute Cost	Dispute Overhead (%)
10	___	___	___	___
50	___	___	___	___
100	___	___	___	___
500	___	___	___	___
1000	___	___	___	___

Tabel 4.4e — Settlement: At Close Only, Dispute: Dispute at Close

TX Count	Gas No-Dispute	Gas w/ Dispute	Dispute Cost	Dispute Overhead (%)
10	___	___	___	___
50	___	___	___	___

100	___	___	___	___
500	___	___	___	___
1000	___	___	___	___

Tabel 4.4f — Fixed Costs (diukur sekali)

Operasi	Gas Cost	Catatan
Channel Open	___	Deploy + initial deposit
Channel Close (cooperative)	___	Both parties agree
Channel Close (unilateral)	___	One party timeout
Dispute Initiation	___	Submit challenge
Dispute Resolution	___	Submit counter-evidence
Settlement (per batch)	___	On-chain state update

4.5 Uji Statistik

- Linear regression: total gas = a × tx_count + b per mode. Report slope (marginal cost per TX).
- Break-even: solve tx_count dimana gas_channel < gas_on_chain. Report exact point + CI.
- ANCOVA: gas = f(mode, tx_count) dengan tx_count sebagai covariate.
- Paired comparison: dispute vs no-dispute pada tx_count yang sama.

4.6 Visualisasi

- Line chart: total gas vs TX count (curves: on-chain, channel close-only, channel every-10, channel every-50, multi-hop). Mark break-even points.
- Bar chart: dispute cost as percentage of total, per TX count.
- Stacked bar: gas breakdown (open + off-chain + settlements + close) per settlement strategy.

TOPIK 3D

Evaluasi Merkle Tree Variants (Standard, Sparse, Verkle, Patricia, Poseidon) untuk State Proof Efficiency

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1. Deskripsi

Implementasi dan benchmarking 5 varian Merkle tree dari scratch di Python. Evaluasi proof size, generation/verification time, storage, dan batch proof pada berbagai state size.

2. Infrastruktur

- Python + hashlib + py_ecc + poseidon-hash. Colab gratis. timeit + tracemalloc. In-memory data.

3. Langkah

Pekan 1: Standard + Sparse. Pekan 2: Verkle + Patricia + Poseidon. Pekan 3: Benchmark harness. Pekan 4: Full benchmarking + report.

4. Skenario Uji Coba

4.1 Variabel Tetap

- SHA-256 (default hash), key 32 bytes, value 32 bytes, Verkle branching 256, seed tetap, GC disabled, 10 warm-up ops.

4.2 Variabel Manipulasi

Variabel	Level
Tree Variant	Standard Merkle, Sparse Merkle, Verkle, Patricia Trie, Poseidon-Merkle
State Size	1K, 10K, 100K, 1M entries
Operasi	Build tree, Generate proof (1 entry), Verify proof, Update 1 entry, Batch proof (100 entries)

Total: 5 tree × 4 state size × 5 operasi = 100 skenario × 5 rep = 500 run. Per run <5 detik. Total: ~2 jam.

4.3 Variabel Respon

Variabel	Satuan	Pengukuran
Build Time	ms	Waktu bangun tree dari scratch
Proof Generation Time	ms	Generate 1 membership proof
Proof Verification Time	ms	Verify 1 proof
Proof Size	bytes	Ukuran proof dalam bytes
Tree Storage Size	MB	Memory untuk tree (tracemalloc)
Update Time	ms	Update 1 entry + recompute root

Batch Proof Size	bytes	Proof untuk 100 entries sekaligus
Batch Verify Time	ms	Verify batch proof

4.4 Template Pengumpulan Data

Tabel 4.4a — State Size: 1K entries

Metrik	Standard	Sparse	Verkle	Patricia	Poseidon
Build Time (ms)	___±___	___±___	___±___	___±___	___±___
Proof Gen (ms)	___±___	___±___	___±___	___±___	___±___
Proof Verify (ms)	___±___	___±___	___±___	___±___	___±___
Proof Size (bytes)	___	___	___	___	___
Storage (MB)	___±___	___±___	___±___	___±___	___±___
Update (ms)	___±___	___±___	___±___	___±___	___±___
Batch Proof Size (bytes)	___	___	___	___	___
Batch Verify (ms)	___±___	___±___	___±___	___±___	___±___

Tabel 4.4b — State Size: 10K entries

Metrik	Standard	Sparse	Verkle	Patricia	Poseidon
Build Time (ms)	___±___	___±___	___±___	___±___	___±___
Proof Gen (ms)	___±___	___±___	___±___	___±___	___±___
Proof Verify (ms)	___±___	___±___	___±___	___±___	___±___
Proof Size (bytes)	___	___	___	___	___
Storage (MB)	___±___	___±___	___±___	___±___	___±___
Update (ms)	___±___	___±___	___±___	___±___	___±___
Batch Proof Size (bytes)	___	___	___	___	___
Batch Verify (ms)	___±___	___±___	___±___	___±___	___±___

Tabel 4.4c — State Size: 100K entries

Metrik	Standard	Sparse	Verkle	Patricia	Poseidon
Build Time (ms)	___±___	___±___	___±___	___±___	___±___
Proof Gen (ms)	___±___	___±___	___±___	___±___	___±___
Proof Verify (ms)	___±___	___±___	___±___	___±___	___±___
Proof Size (bytes)	___	___	___	___	___
Storage (MB)	___±___	___±___	___±___	___±___	___±___
Update (ms)	___±___	___±___	___±___	___±___	___±___
Batch Proof Size (bytes)	___	___	___	___	___
Batch Verify (ms)	___±___	___±___	___±___	___±___	___±___

Tabel 4.4d — State Size: 1M entries

Metrik	Standard	Sparse	Verkle	Patricia	Poseidon
Build Time (ms)	___±___	___±___	___±___	___±___	___±___
Proof Gen (ms)	___±___	___±___	___±___	___±___	___±___

Proof Verify (ms)	___±___	___±___	___±___	___±___	___±___
Proof Size (bytes)	___	___	___	___	___
Storage (MB)	___±___	___±___	___±___	___±___	___±___
Update (ms)	___±___	___±___	___±___	___±___	___±___
Batch Proof Size (bytes)	___	___	___	___	___
Batch Verify (ms)	___±___	___±___	___±___	___±___	___±___

4.5 Uji Statistik

- Two-way ANOVA: tree_variant × state_size pada setiap metrik.
- Log-linear regression: proof_size = a × log(state_size) + b per variant — validasi theoretical O(log n).
- Friedman test: 5 variants pada metrik yang sama.
- Post-hoc Nemenyi: identifikasi pasangan signifikan.

4.6 Visualisasi

- Line chart: proof size vs state size per variant (log-log scale).
- Line chart: build time vs state size per variant (log scale x-axis).
- Grouped bar chart: proof gen/verify time per variant pada 100K entries.
- Bar chart: batch proof size advantage (Verkle vs others) per state size.